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A LANGUAGE BASED ON THE FUNDAMENTAL FACTS OF SCIENCE

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The problem of how to communicate with the members of an alien society has been discussed by many authors but only one, Hans Freudenthal, has constructed a language for this purpose. Freudenthal assumes nothing other than the ability to reason as humans do and, because he assumes so little, it is necessary to communicate a great deal about the language itself before being able to communicate any interesting information. The problem is here approached differently. Since it is likely that contact between our civilization and an alien one would be *via* radio, potential correspondents would have a basic knowledge of science. Such beings should therefore be able to learn a language based on fundamental science. It is assumed, more specifically, that our correspondents can count, understand chemical elements, are familiar with the melting and boiling behaviour of a pure substance and understand the properties of the gaseous state. All this should be known to any society capable of developing the radio telescope. By systematically using this common knowledge one can communicate notation for numbers and chemical elements and then communicate our basic physical units; i.e., the gram, the calorie, the degree (Kelvin), etc. Once this is done more interesting information can be exchanged.

1. INTRODUCTION

The purpose of this paper is to show how a language can be developed, based on the rudiments of logic, mathematics and physical sciences which would enable us to communicate with hypothetical intelligent aliens. There are, of course, other reasons for constructing such a language. First, in the not very distant future, we shall want to involve the computer in scientific problems at a level akin to that of humans. This will require a language which incorporates science into its basic structure. Natural languages, i.e., any language spoken by human beings, are too complicated for this and are unsuitable in other ways as well. Secondly, it is necessary to elucidate the manner in which real-world experience enters into language acquisition. This information could be valuable in the development of artificial intelligence, particularly in the area of machine translation. In developing a knowledge-based system to translate material in a certain area, it is important to know what minimal knowledge the system must have.

The idea of constructing a logic-based language goes back at least as far a Leibnitz (1646-1716). Late in the nineteenth century Peano (1858-1932) discussed such languages extensively and his ideas strongly influenced Russell and Whitehead [3]. These efforts had an important impact on the development of mathematical logic and on the foundations of mathematics [3]. Computer scientists, in discussions of the problems encountered in the creation of programming languages, drew on this work [1]. However, the idea of using logic-based languages as a means of communication, the original intent of both Leibnitz and Peano, was forgotten until 1960. In this year, H. Freudenthal published a remarkable book [4] called Lincos (a contraction of the words Linguia Cosmica) wherein he set himself the problem of designing a language suitable for communicating via radio with (hypothetical) intelligent extra-terrestrials. Since such beings would have no knowledge of our natural languages and could not be shown physical objects or demonstrations, there is nothing but logic on which to base a common language. Freudenthal showed that, if their thought processes are human-like, a language suitable for mutual communication could be taught to them.

Although Freudenthal had his critics, he also had supporters [4]. His work attracted the attention of computer scientists [1] and of those interested in SETI (the search for extraterrestrial intelligence) [6,8]. A different situation occurs if it is assumed that ET's have a technology for this implies a detailed knowledge of the physical universe. We suppose that they have investigated certain basic scientific problems; e.g., the fundamental properties of matter and of energy. How does one measure these things and how do they interact? An alien science may be radically different from our own but when intelligent races describe the same phenomena, particularly uncomplicated chemical or physical processes, the descriptions must be equivalent in the sense that they both really describe the same phenomenon. Hence, it does not seem unreasonable that these descriptions would be mutually understandable.

This additional assumption enables us to move rapidly to the point where interesting information can be exchanged. Communication of this kind must begin with modest goals. The best one can hope for is to develop the facility to exchange precise information. It may be easy to devise some scheme for saying that we live on a planet but not very informative. To be able to state that the planet has a specified mass, radius, atmosphere, etc. is to convey precise and useful information. This goal is different from Freudenthal's aim of constructing a complete language. The advantage of adopting a technical language is that exchanging valuable information can begin before the language is complete. Information is exchanged as the language is developed, the information received helps the user to carry the language further, whereas Freudenthal's Lincos is an actual language which contains words for many common concepts which must be learned as they are presented to be able to read further.

We have assumed that radio waves would be used and that the aliens have devices for sending and receiving such waves, while Freudenthal goes so far as to suggest that they might have sense organs for this purpose; at least for receiving. Many argue that the sense of sight must be universal and so television or picture messages should be sent [5]. On the other hand, even given the sense of sight, there is no way of knowing how an alien would interpret a picture. An ingenious method, to counter this argument, using π and the properties of circles has been devised for establishing a television "rastor" at the start of such an exchange [6]. There is no need to make an assumption one way or the other regarding the sense of sight though there is no doubt that its presence would simplify matters in many respects. For example, a picture of a balance scale with weights does not convey the gram to someone unaware of this unit; it merely shows, if understood, that the two weights are of the same or of different mass. So communical via television would change some of our problems and simplify others, but would not eliminate them.

2. FORMAT

The proposed technically based language may be presented as a series of stages rather than as dialogue, with the content of each stage dictated by prior stages. It turns out to be easy, once begun, to continue discussing mathematics or, again once begun, chemistry. The chief difficulty lies in going from a discussion of mathematics to a discussion of chemistry and from there to a discussion of physics, so certain scientific facts have been used to provide "links" between these subjects. They are either distinctive phenomena, like the melting and boiling of a pure substance, or fundamental, like the notion of atomic weights which enables us to do meaningful chemical calculations. Standard symbols for numbers, chemical elements, etc. have been used to enhance readability. Furthermore, any statement made in symbols is followed by a translation into English.

The attempt at communication may begin by assuming that the aliens are familiar with the process of counting. The rudiments of formal logic can then be developed. The basic symbols of this subject are the connectives: "and" denoted by $\wedge$, "or" denoted by $\vee$, "not" denoted by $\sim$, "implies" (the "if-then" of programming languages) denoted by $\rightarrow$, and "logical equivalence" denoted by $\leftrightarrow$. To illustrate the use of these symbols suppose that n is a natural number (i.e., $n = 1, 2, 3, 4, 5, \ldots$), thenwe may write: $(n < 2) \rightarrow (n = 1)$. In words: "if a natural number is less than 2 then it is the natural number 1." Another example $(n < 3) \rightarrow (n = 1) \vee (n = 2)$. In words. "if a natural number is less than 3 then it is either 1 or 2."

Greater flexibility is added to the language by introducing logical quantifers: $\forall$ read "for all" and $\exists$ read "there is". Using these we may write, for example, $\forall\ (n)\ (n \leq n^2)$ and $(\exists\ n)\ (n = n^2)$. In words the first of these reads "All natural numbers are less than or equal to their squares," while the second reads "There is a natural number which is equal to its square."

Next is the notation of set theory. Informally, a set is just a collection of any objects e.g., the collection of all leaves on a certain tree is a set and, of course, this is not a leaf but a new object with properties all its own. Similarly, a flock of sheep is a set which has propreties and characteristics of its own, different from those of an individual sheep. When possible, the objects forming a set are listed between two curly brackets. So the set consisting of the two letters a and b is written $\{a,b\}$, and the set of all natural numbers is written $\{1, 2, 3, 4, \ldots\}$; the latter set often denoted simply by N. The symbol $\in$ is used to say that a particular object is in a set. So $a \in \{a,b\}$, $b \in \{a,b\}$, $1 \in N$, $6 \in N$, $23 \in N$, etc. To say that an object is not in the set we use $\notin$. So $c \notin \{a,b\}$, $5 \notin \{a,b\}$, $\frac{1}{2} \notin N$, $\pi \notin N$, etc. It is sometimes necessary to separate out from a given set those objects in the set which have a certain property. This gives us a new set. For example, the set of natural numbers greater than 5 may be written $\{n \in N \mid n > 5\}$. This is read "the set (that is the way curly brackets are read) of all natural numbers n such that (this is the meaning of the vertical bar) $n > 5$. One last example. When F is a flock of sheep and it is necessary to collect together into a new set those that weigh more than 30 pounds, we write $\{s \in F \mid s \text{ weighs more than 30 pounds}\}$.

3. STAGE 1 - THE BEGINNINGS OF A LANGUAGE

Assuming that we are in radio communication with an alien race alone, implies that they have a sophisticated technology. The first assumption about their intellectual abilities will be that they can count and that the mental constructs which we call the natural numbers (i.e. 1, 2, 3, 4, ...) are understood. This collection has obvious properties which we may assume are also known. Specifically, the set is totally ordered and (countably) infinite. Counting is nothing more than setting up a one-to-one correspondence between a collection of objects, the things to be counted, and the first, say, n integers. Hence we may assume that the function concept is known, or at least one kind of function is known. Finally, it is supposed that the two shortcuts to counting - addition and multiplication - have been discovered. The idea of equality is implicit in this last assumption. For instance, one plus two is the "same thing" as three.

The attempt at communication may thus begin by presenting the notation for these shared concepts in a simple repetitive way. This is similar to what Hans Freudenthal did in his book Lincos [3]. It is only in stage three, where the assumption that aliens have a knowledge of science is introduced, that the approach diverges. We take "." to be an iconic representation of the natural number 1. The dot is our symbol for, say, a single beep in an actual transmission. Next 2 is represented by ". ." where the space between the beeps is a pause of "unit" length. It must be supposed that listeners will understand that transmissions represent different entities when they are separated by time. A comma is used to represent a pause of two units length, two commas, a pause of four units length and so forth. The phrase "and so on" is not part of the transmission. it simply means that other examples of the type given should, or may, be sent. This is a concession to terrestrial readers.

. ,, .. ,, ... ,, ... ,, ... ,, (1.1)

=,, =,, =,, ,2,3,4,5,6,7,8,9..
1,2,3,4,5,6,7,8,9.. (1.2)

.=1 ,, 1=.,, .. = 2,, 2= .. (1.3)

Numbers are written in base 10 notation only because this is convenient. Base 2 might be more efficient in an actual transmission.

+,, +,, +,, .+. + ..,,.+.=2,,1+1=2,,.+..=..+.=...,,..+.
=3=.+..,,2+1=3=1+2,, (1.4)

x,,x,,x,,.x.=.,,.x.=1,,1x1=1,, and so on. (1.5)

Arithmetic is used to present the enumeration scheme for the natural numbers.

$$0,,0,,10,,9+1=1+9=10,,\ 2x5=5x2=10,,\ 11,11,10+1$$
$$=1+10=11,,\ 12,,12,,\ 10+2=2+10$$
$$=12,,\ 2x6=6\ x\ 2=12,,\ 4\ x\ 3=3\ x\ 4=12,, \tag{1.6}$$

$$1,2,3,4,5,\text{etc},,,\ 1,2,3,4,5,6,7,7,9,10,11,12,\text{etc.}\quad \text{and so on.} \tag{1.7}$$

The symbol "etc" is a way of indicating that the natural numbers comprise an infinite set. This symbol will be useful in other connections.

$$<,,<,,<,,1<2,,2<3,,3<4,,4<5,,\text{etc}$$
$$1<2,,2<3,,1<3,,,,3<4,,2<4,,1<4,,,$$
$$4<5,,3<5,,2<5,,1<5,,\text{etc}\qquad \text{and so on.} \tag{1.8}$$

The natural numbers, their arithmetic and their order are now shared concepts. A common notation for these concepts is also shared. This tiny mutual vocabulary may now be used to develop a simple language, beginning with the concept of a variable and a punctuation device.

$$a,,b,,a,,b,,a,,b,, \tag{1.9}$$

$$1+2=2+1,,\ 1+3=3+1,,\text{etc},,$$
$$1+a=a+1,,\ 2+3=3+2,,\ 2+4=4+2,,\text{etc},,$$
$$2+a=a+2,,,,1+a=a+1,,2+a=a+2,,3+a=a+3,,\text{etc},,$$
$$b+a=a+ \tag{1.10}$$

$$(,,),,,,(,,),,,,(,,),,,$$
$$a+a=2a,,\ a+a+a=a+(a+a)=a+2a=3a$$
$$a+a+a+a=(a+a)+(a+a)=2a+2a=4a$$
$$a,,b,,c,,a+b+c=a+(b+c)=(a+b)+c$$
$$axbxc=ax(bxc)=(axb)xc$$
$$ax(b+c)=(axb)+(axc)\qquad \text{and so on.} \tag{1.11}$$

The logical connectives may now be introduced: implies ($\rightarrow$), logical equivalence ($\leftrightarrow$), negation ($\sim$), or ($\vee$) and, finally, ($\wedge$).

$$(a+1=2)\rightarrow (a=1),,\ (a=1)\rightarrow (a+1=2),,\ (a+1=2)\leftrightarrow (a=1),,$$
$$(2<a)\rightarrow(1<a),.\sim((1<a)\rightarrow(2<a))$$
$$(a<2)\leftrightarrow(a=1),,(a<3)\leftrightarrow((a=1)\vee(a=2))$$
$$(a=2)\rightarrow(a<3),,(a=2)\rightarrow(1<a),,(a=2)\leftrightarrow((a<3)\wedge(1<a)) \tag{1.12}$$

$$(a\le b)\leftrightarrow(a=b)\vee(a<b),,\ (a=b)\leftrightarrow(a\le b)\wedge(b\le a),,$$
$$(a\ne b)\leftrightarrow\sim(a=b),,\ (a\ne b)\leftrightarrow((a<b)\vee(b<a)).\ \text{and so on} \tag{1.13}$$

This last line simply defines the useful mathematical symbols $\le$ and $\ne$.

It is now assumed that the aliens can mentally aggregate the members of a collection of distinct objects into a whole or totality, thereby obtaining a new entity. This capability is built into the Pascal programming language and even very primitive societies recognize that a herd or flock of animals is different from an individual animal. Informal mathematical terminology will be used to call any collection of objects a set. Some elementary set theory will greatly enhance the descriptive powers of the language and we are still close to the process of counting. To count some objects we mentally separate them from all other objects thereby collecting them into a set, and the process of counting tells us the cardinality (numbers of members) of this set.

The elements or members of a set are usually listed between two curly brackets. So the set formed from the objects a and b is denoted by $\{a,b\}$. The relationship between an object and a set is expressed by the symbol $\in$. We write $a \in S$ if a is a member of the set S and $a \notin S$ otherwise.

$$1,,\ \{1\},,\sim(1=\{1\}),,\ 1\in\{1\},,\ (a\in\{1\})\leftrightarrow(a=1),,1,2,,$$
$$\{1,2\},,1\in\{1,2\},,\ 2\in\{1,2\},,\ (a\in\{1,2\})\leftrightarrow(a=1)\vee(a=2),,$$
$$(\{a,b\}=\{1,2\})\leftrightarrow((a=1)\wedge(b=2))\vee((a=2)\wedge(b=1)),,$$
$$\{1,2\}=\{2,1\}. \tag{1.14}$$

Two sets are defined as equal if they have exactly the same members. The same symbol for equality of sets is used as for an equality of numbers. Using the same symbol for "slightly" different things does not cause any misunderstanding among people, but in a project of this kind we should be wary of such practices.

It is now necessary to develop some of the basic notions of set theory by making use of the infinite set of natural numbers. There may be some question about the wisdom of using infinite sets. One can probably avoid them for the first few stages but there is no way for humans to avoid infinity, for infinite processes, in one form or another, underlie most fundamental mathematical models.

$$N,,N,,N=\{1,2,3,\text{etc}\},,\quad N=\{1,2,3,4,5,\text{etc}\},,$$
$$(n\in N)\leftrightarrow(n=1)\vee(n=2)\vee(n=3)\ \text{etc},,$$
$$N,,N,,O,,E,,P,,E=\{2,4,6,\text{etc}\},,\ O=\{1,2,3,\text{etc}\}$$
$$E=\{2,4,6,8,10,\text{etc}\},,$$
$$(n\in N)\leftrightarrow(n=2)\vee(n=4)\vee(n=6)\vee(n=8)\vee\text{etc}$$
$$(n\in O)\leftrightarrow(n\in N)\wedge(\sim(n\in E)),,\ O=\{1,3,5,\text{etc}\}$$
$$P=\{2,3,5,7,11,13,17,19,\text{etc}\}. \tag{1.15}$$

This last set is, of course, the set of prime numbers. The importance of these numbers has long been recognised by humans and may be known to other races as well. To indicate that something is a set we list it with known sets.

$$N,E,O,P,A,B,,$$
$$(n\in E)\rightarrow(n\in N),,\ E\subseteq N,,\ O\subseteq N,,\sim(O\subseteq E),,\sim(E\subseteq O),,$$
$$(A\subseteq B)\leftrightarrow((a\in A)\rightarrow(a\in B)),,\ (A=B)\leftrightarrow(A\subseteq B)\wedge(B\subseteq A)$$
$$(a\in A\cup B)\leftrightarrow(a\in A)\vee(a\in B),,\ O\cup E=N$$
$$(a\in A\cap B)\leftrightarrow(a\in A)\wedge(a\in B),,\ O\cap N=O,,E\cap N=E,\ O$$
$$\cap E=\varnothing \tag{1.16}$$

We turn now to a more flexible notation for sets.

$$\{,,\},,,,\{,,|,,\},,,,\{,,|,,\},,$$
$$\{1\}=\{n\in N|n=1\},,,\{1,2\}=\{n\in N|n\le 2\},,$$
$$\{2,4\}=\{n\in E|n\le 5\},,\ \{1,3,5\}=\{n\in O|n\le 5\}$$
$$p\in N,,q\in N,,\ (\{n\in N|n\le p\}\subseteq\{n\in N|n\le q\}\leftrightarrow(p\le q)). \tag{1.17}$$

The symbol $\{...|---\}$ is read "the set of all ... such that ---." We may now introduce the quantifiers $\forall$, read "for all," and $\exists$, read "there is," into our language.

$$(m\in N)\rightarrow(2\ x\ m\in E),,\ (\forall m\in N)(2\ x\ m\in E)$$
$$(m\in N)\rightarrow((2\ x\ m+1)\in O),,\ (\forall m\in N)((2\ x\ m+1)\in O)$$
$$(n\in E)\rightarrow(\exists\, m\in N)(n=2\ x\ m),,\ (\forall n\in E)(\exists\, m\in N)(n=2\ x\ m)$$
$$(n\in O)\wedge(1<n)\rightarrow(\exists\, m\in N)(n=2\ x\ m+1),,$$
$$\text{and so on.} \tag{1.18}$$

4. STAGE 2 - MATHEMATICS

The language is now rich enough to enable us to include some non-trivial things. Any technological society must come to terms with fractions, with zero and the negative numbers, and with the fact that very large and very small numbers arise in the sciences.

Discussing these things serves two purposes. First, it gives a common notation for these very useful concepts. Secondly, since these simple extensions of the natural numbers should be well-known and easily recognised, it provides the aliens with a check on their understanding of our logico-linguistic symbols.

$a \in N, b \in N,, (x+a=b) \wedge (a<b) \rightarrow x \in N,, (x+a=b) \wedge (a \geq b) \rightarrow \sim(x \in N)$
$N,Z,N \subseteq Z,, \sim(Z \subseteq N),,(z \in Z)(w \in Z)((z+w \in Z) \wedge (z+w=w+z))$
$0 \in Z,, \sim(0 \in N),, \forall z \in Z)(z+0=0+z=z),,$
$(\forall z \in Z)(\exists(-z) \in Z)\ (z+(-z)=0)$
$(\forall z \in Z)((z \in N) \vee (z=0) \vee (-z) \in N)$
$(a \in Z)(b \in Z)((x+a=b) \rightarrow (x \in Z))$ (2.1)

We have said that certain equations have no solutions in N, that there is an extension Z of N which contains an additive identity, which we identify with the zero used above as a place holder, and which contains an additive inverse for each of its elements. We note that every element of z is either zero, in N or has an additive inverse in N and, finally, we note that the equations we couldn't always solve in N can always be solved in Z.

$(\forall z \in Z)(\forall w \in Z)\ ((z \times w \in Z) \wedge (z \times w = w \times z))$
$1 \in N,,N \subseteq Z,,1 \in Z,,(\forall z \in Z)(z \times 1 = z)$
$(\forall z \in Z)(\forall w \in z)(\forall y \in Z)((z+w)+y =$
$z+(w+y)) \wedge ((z \times w) \times y = z \times (w \times y))$
$(\forall z \in Z)(\forall w \in Z)(\forall y \in Z)(z \times (w+y) = z \times w + z \times y).$ (2.2)

Line 2.2. gives the multiplicative structure of Z and relates it to the additive structure. Note that we have not constructed Z, merely asserted its existence. An equally pragmatic view will be taken of the set Q of rational numbers, but here we shall at least indicate how this set may be constructed.

$2x=3, \sim(x \in Z),, 5x=3, \sim(x \in Z)$
$Z,Z, Q,,Q,\langle a,b\rangle \in Q \leftrightarrow (b \in Z) \wedge (a \in Z) \wedge (b \neq 0) \wedge (bx = a)$
$Q = \{\langle a,b\rangle \mid a \in Z, b \in Z, b \neq 0\},, \langle a,b\rangle = a/b.$ (2.3)

We give some equations whose coefficients are integers but whose solutions are not integers. We then state that Q consist of pairs of integers related via an equation: so $\langle a,b\rangle$ is the solution to $bx = a, b \neq 0$. This last statement, $\langle a,b\rangle = a/b$, is made for the convenience of terrestrial readers. This familiar notation is used now.

$(\forall a/b \in Q)(\forall c/d \in Q)((a/b = c/d) \leftrightarrow (ad = bc))$
$(\forall a/b \in Q)(\forall c/d \in Q)(a/b + c/d = ad + bc/bd)$
$(\forall a/b \in Q)(\forall c/d \in Q)(a/b \times c/d = a \times c/b \times d)$
$(\forall z \in Z)(z/1 \in Q),, z = z/1,, Z \subseteq Q$
$(\forall b \in Q)(\forall a \in Q)((b \neq 0) \wedge (bx = a) \rightarrow x \in Q).$ (2.4)

This last line defines the equality and arithmetic operations on the set Q. It gives our formal identification of $z \in Z$ with z/1 $\in Q$, hence we may regard Z as a subset of Q. Finally, equations of the kind we started with, which need not have integer solutions, always have rational solutions. Notice that the concept of ordered pair has been more or less defined.

$(\forall q \in Q)(q^1 = q),, (\forall q \in Q)(\forall n \in N)(n \neq 1)(q^n = q \times q^{n-1})$
$(10)^1 = 10,, (10)^2 = 100,, (10)^3 = 1000,,$ etc
$(\forall q \in Q)(q \neq 0)(q^0 = 1),,(\forall q \in Q)(q \neq 0)(n \in N)(q^{-n} = 1/q^n)$
$(10)^0 = 1,, (10)^{-1} = 1/10 = 0.1,, (10)^{-2} = 1/10^2 = 0.01,$ etc.
$602252 \times 10^{19} = 60225.2 \times 10^{20} = 6022.52 \times 10^{21} =$ etc $= 6.02252 \times 10^{23}$
$83.434 \times 10^{-5} = 83143.4 \times 10^{-4} = 8314.34 \times 10^{-3} =$ etc $= 8.31434.$
(2.5)

Line 2.5 just gives our useful exponential notation. This stage can be concluded with the function concept, which takes us back to counting.

$a,,b,,\{a\},,\{b\},,\{a,b\},,\{\{a\},\{a,b\}\}$
$\langle a,b\rangle,, \sim(\langle a,b\rangle = \{a,b\}),,\langle a,b\rangle = \{\{a\},\{a,b\}\}$
$(\{u,v\} = \{a,b\}) \leftrightarrow (((u=a) \wedge (v=b)) \vee ((u=b) \wedge (v=a)))$
$(\langle u,v\rangle = \langle a,b\rangle) \leftrightarrow (u=a) \wedge (v=b).$ (2.6)

In the line above, we gave a formal definition of the notion of ordered pair. The reader need only recall that $\langle 1,2\rangle$ represents a point in the plane and $\langle 2,1\rangle$ represents another, quite different, point. The familiar plane of elementary mathematics is just the set of all ordered pairs of numbers, and we have just seen that the rational numbers are ordered pairs of integers (subject to certain conditions). All functions dealt with are also sets of ordered pairs even though rarely looked at in this way. However, when we need to define function, as we now do, it is useful to fall back on the concept of ordered pair.

$N,Z,Q,S,T,,S \times T$
$S \times T = \{\langle s,t\rangle \mid s \in S, t \in T\}$
$((s \in S)(t \in T)(\langle s,t\rangle \in S \times T)) \wedge ((\forall u \in S \times T)(\exists s \in S)(\exists t \in T)$
$(u = \langle s,t\rangle))$ (2.7)

Line 2.7 defines the Cartesian product of the sets S and T. The plane is just the Cartesian product of the x-axis with the y-axis.

We now give some examples of functions using counting and then an abstract definition.

$\{1,2,3\} \subseteq N,, \{a,b.c\},,$ count $= \{\langle 1,a\rangle,\langle 2,b\rangle,\langle 3,c\rangle\},,$ count $\subseteq \{1,2,3\} \times \{a,b,c\},,$ count $(1) = a,,$ count $(2) = b,,$ count $(3) = c,,,,$ card $\{a,b,c\} = 3$ and so on. (2.8)

Note that we have defined the cardinality of a finite set. We finish up now with more general functions.

$N,Z,Q,,S,T,,S \times T,, \mathrm{Fun}(S,T),,$
$f \in \mathrm{Fun}(S,T) \rightarrow (f \subseteq S \times T)(f \neq \varnothing),,,,f \in \mathrm{Fun}(S,T),,(f(s) = t) \leftrightarrow (\langle s,t\rangle \in f)$
$(f \in \mathrm{Fun}(S,T)) \leftrightarrow (\forall s \in T)(f(s) = t) \wedge (\forall u \in S)(\forall v \in S)((u = v) \rightarrow$
$(f(u) = f(v))).$ (2.9)

It would be easy to continue with mathematics, but our purpose is to develop a language suitable for general scientific use. We shall not even discuss the real numbers even though many physical qualities are, theoretically, real valued. However, the result of any actual measurement is a rational member and so we shall define mass, temperature, etc., to be rational-valued functions.

5. STAGE 3 - MATTER

We now need to begin discussing the physical universe. The starting point is the set of chemical elements for it is likely that any society with the capacity to send and receive radio messages must be familiar with basic chemistry. The difficulty is not that they won't know about the elements, but rather in getting them to understand that the elements are what we are talking about. This is why a number of different clues are given that perhaps individually, but certainly collectively, make clear what is meant.

The first two stages gave the notation for the natural numbers N (1.6 and 1.7), for the rational numbers Q (2.7) and for functions (2.11). The meaning of $\in$ ("belongs to" 1.16), of $\subseteq$ ("is contained in" 1.8) and of the Cartesian product of two sets (2.6). We also discussed the set notation {...|---} ("the set of all ... such that ---" 1.19) and the cardinality (i.e., number of elements 2.8) of finite sets.

$$N,,Z,,O,,P,Q,,At$$
$$(x \in At) \leftrightarrow (x = 1 \text{ atom}). \tag{3.1}$$

To introduce a new set, At, we list it with known sets as before, and give a name to the elements of At. This is the first time this has been done.

$$Num \in Fun(AT,N) \tag{3.2}$$

This simply says that there is a function Num from At into the natural numbers; that is, to each atom x a positive whole number Num (x) is attached. This function cannot be specified further but will be used in such a way that its meaning will eventually become clear. Of course, Num(x) is the atomic number of x.

$$H = \{x \in At|Num(x) = 1\},, He = \{x \in At|Num(x) = 2\},$$
$$\text{and so on} \tag{3.3}$$

In line 3.3. hydrogen is defined to be the set of all atoms having atomic number 1, helium as the set of all atoms having atomic number 2, and so forth. The meaning of lines 3.2 and 3.3 will probably not be clear to an alien, but the chemical elements also possess certain natural periodicities, conveniently expressed in the periodic table.

$$Elm = \{H,He,Li, ..., U\}, card(Elm) = 92$$
$$T=\{1,2,3,4,5,6,7\}x\{1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18\}$$
$$Per \in Fun(T,Elm \cup \{\varnothing\}). \tag{3.4}$$

We have given the name Elm to the set of naturally occurring elements and we give the cardinality of this set. This alone should be suggestive. Next we define a seven by eighteen "checkerboard" T and mention of function Per from T into Elm together with the empty set. This may seem a little pedantic, but should add clarity when the function Per is defined. Incidentally, terrestrials will want to read "⟨1,2⟩" as "row one, column two," and so forth.

$$Per (\langle 1,1\rangle) = H,, Per(\langle 2,1\rangle) = Li,, Per (\langle 3,1\rangle) = Na,,$$
$$Per (\langle 4,1\rangle) = K,, Per(\langle 5,1\rangle) = Rb,, Per (\langle 6,1\rangle) = Co,,$$
$$Per (\langle 7,1\rangle) = Fr,,,,. \tag{3.5}$$

Here the first column of the periodic table has been given in the hope that this will be more easily understood than the first row. There are places in our seven by eighteen checkerboard where no element occurs and at any such location we define Per to be the empty set. This is consistent with our definition of an element as a particular set.

$$Per (\langle 1,2\rangle) = \varnothing,, Per (\langle 2,2\rangle) = Be,, Per (\langle 3,2\rangle) = Mg,,$$
$$Per (\langle 4,2\rangle) = Ca,, Per (\langle 5,2\rangle) = Sr,,$$
$$Per (\langle 6,2\rangle) = Ba,, Per (\langle 7,2\rangle) = Ra \quad \text{and so on} \tag{3.6}$$

An alien "chemist" faced with a set of about 100 objects each assigned a positive integer would, at least tentatively, assume that the chemical elements are being described. If that is coupled with the tabular, periodic, array presented next, it is very probable that our "chemist" would conclude that the elements are being discussed.

The specific meaning of these symbols can be reinforced by calling attention to the fact that some atoms combine. This requires a little more terminology.

$$x \in At,, (x = 1 \text{ atom},C) \leftrightarrow (x \in At \cap C)$$
$$1 \text{ atom, He} + 1 \text{ atom, He} = 2 \text{ atom, He}$$
$$1 \text{ atom, Kr} + 1 \text{ atom, Kr} = 2 \text{ atom, Kr}$$
$$1 \text{ atom, H} + 1 \text{ atom, H} \rightarrow 1m, H,,, 1m, H = H_2$$
$$1 \text{ atom, O} + 1 \text{ atom, O} \rightarrow 1m, O,,, 1m, O = O_2$$
$$2 \text{ atom, H} + 1 \text{ atom, O } 1m, H_2O$$
$$1 \text{ atom, C} + 2 \text{ atom, O} \rightarrow 1m, CO_2$$
$$1 \text{ atom, N} + 3 \text{ atom, H} \rightarrow 1m, NH_3 \quad \text{and so on} \tag{3.7}$$

Line 3.7 gives some terminology first, then situations where the atoms do not react and, finally, a number of common instances where combination does take place to produce something new. The symbol → has been used to indicate chemical change and this is ambiguous since → already means "implies". Such ambiguities do not trouble humans because they are aware of the context but they might be confusing in a project such as this one. Perhaps other notations should be devised instead.

In building a language for communicating with an alien society we will, inevitably, be saying quite a lot about ourselves. The discussion up to this point, however, has been strictly impersonal. To start the process of humanisation we need to communicate the meaning of our basic units of measurement: the gram, the degree (Kelvin), our units of volume, etc. These give a rough idea of our size and a very good idea of how we "see" and think; how the human mind brings some order to the complex picture presented to us, and the aliens, by the physical universe. This is why, besides the fact that we think that the chemical elements and the periodic table would be universally recognised, we began this message with a discussion of chemistry. Chemists everywhere are faced with the problem of relating their work in the laboratory, where "large" quantities of matter interact, with the changes taking place on the molecular level.Human chemists solved this problem by introducing a system of atomic weights and using the Avogadro number. Alien chemists must also solve this problem so it is reasonable to assume that they are familiar with this number; expressed, of course, in their units. If this is so, then a simple calculation based on the information below will enable them to convert our gram into units which they understand.

We begin by presenting our system of atomic weights. This is based on assigning an atomic weight of 12 to a particular isotope of carbon. For the sake of brevity we have not treated isotopes here, but our understanding of them is implied by the fact that many atoms have non-integer atomic weights.

$$Q,,At,,amas \in Fun(At,Q)$$
$$amas(1 \text{ atom,C}) = 12,,amas(1 \text{ atom, H}) = 1.008$$
$$amas(1 \text{ atom, C}) = 12,,amas(1 \text{ atom,O}) = 16$$
$$amas(1 \text{ atom,C}) = 12,,amas(1 \text{ atom,N}) = 14.008 \tag{3.8}$$

We now extend the domain of our function amas

$$amas(1m,O) = amas(O_2) = amas(1 \text{ atom,O}) + amas(1 \text{ atom,O}) = 32$$
$$amas(1m,H) = amas(H_2) = amas(1 \text{ atom,H}) + amas(1 \text{ atom,H}) = 2.016$$
$$amas(1m,H_2O) = amas(H_2) + amas(1 \text{ atom,O}) = 18.016$$
$$amas(1m,CO_2) = amas(1 \text{ atom,C}) + amas(O_2) = 44$$
$$\text{and so on} \tag{3.9}$$

We cannot yet give a general definition of matter but can introduce the family of all sets of atoms. We have denoted it by "Cmat", for "common matter" but this terminology merely serves as a mnemonic for terrestrial readers. This set proves useful in lines 3.10, 4.2, 44 and 5.9 but should be viewed as only a temporary first step towards a definition of matter.

$$
\begin{array}{ll}
\text{Cmat} = \{L \mid L \subseteq \text{At}\},, \text{mass} \in \text{Fun}(\text{Cmat}, Q) & \\
(L \in \text{Cmat}) \wedge (L \subseteq C) \wedge (\text{card}(L) = 6.023 \times 10^{23}) \rightarrow (\text{mass}(L) = 12 \text{ gram}) & \\
(L \in \text{Cmat}) \wedge (L \subseteq \text{He}) \wedge (\text{card}(L) = 6.023 \times 10^{23}) \rightarrow (\text{mass}(L) = 4.003 \text{ gram}) & \\
(L = 12\text{g}, C) \leftrightarrow (L \in \text{Cmat}) \wedge (L \subseteq C) \wedge (\text{mass}(L) = 12 \text{ gram}) & \\
\text{and so on} & (3.10)
\end{array}
$$

6. STAGE 4 - ENERGY

The members of any technological society must be familiar with heat, with the fact that heat raises the temperature of a body and the fact that enough heat will change the state of a body. It is assumed that the aliens have devised a temperature scale, a method for measuring heat, and have studied the states of matter and the process of changing from one state to another. In particular, the gaseous state must be understood by any scientific society since failure to understand gases can lead to serious accidents. As far as we know, it was the study of gases which first led humans to the concept of absolute zero. It is assumed that the aliens share this concept with us. Our objective is to communicate our degree (Kelvin), our calorie and the many related concepts mentioned above beginning with the fact that chemical reactions often produce heat and sometimes require heat to get them started.

$$
\begin{array}{ll}
\text{Fe} + \text{S} = \text{Fe} + \text{S},, \text{Fe} + \text{S} + \Delta \rightarrow \text{FeS} & \\
1\text{m}, \text{K Cl O}_3 + \Delta \rightarrow 1\text{m}, \text{K Cl} + 3 \text{ atom}, \text{O} & \\
12\text{g}, \text{C} + 32\text{g}, \text{O} \rightarrow 44\text{g}, \text{CO}_2 + \Delta & \\
12\text{g}, \text{C} + 32\text{g}, \text{O} \rightarrow 44\text{g}, \text{CO}_2 + 94 \times 10^3 \text{ cal}, \Delta \text{ and so on.} & (4.1)
\end{array}
$$

We have said that there is a "quantity" Δ which is produced by certain chemical reactions and which triggers some reactions. This Δ could not represent matter since all matter is accounted for in our equations. Perhaps this alone is enough to communicate the fact that Δ is our way of representing heat. If so, then the last line in 4.1 already gives the alien a meas of finding our calorie. In any case we have tried below to give more clues to the meaning of Δ.

$$
\begin{array}{ll}
\text{Cmat}, Q,, \text{Temp} \in \text{Fun}(\text{Cmat}, Q) & \\
(L \in \text{Cmat}) \wedge (\text{Temp}(L) = 5 \text{ deg}) \leftrightarrow (L \text{ @ } 5 \text{ deg}) & \\
(\forall L \in \text{Cmat})(\text{Temp}(L) \geq 0) & \\
(\forall L \in \text{Cmat}) (((L + \Delta) \in \text{Cmat}) \wedge (\text{Temp}(L) < \text{Temp}(L + \Delta))) & \\
\text{and so on} & (4.2)
\end{array}
$$

We have said that we have a function Temp which is closely connected to this Δ introduced earlier and observed that Temp is non-negative and that adding Δ to matter just increases Temp. Our degree can be found from the information given on the next line.

$$
\begin{array}{ll}
\text{Ag @ } 1232 \text{ deg} = \sigma \text{Ag} & \\
\text{Ag @ } 1233 \text{ deg} = \sigma \text{Ag} & \\
(1\text{g}, \sigma \text{AG @ } 1234 \text{ deg} + 21.1 \text{ cal}, \Delta) = (1\text{g}, \lambda \text{ Ag } 1234 \text{ deg}) & \\
\text{Pb @ } 598 \text{ deg} = \sigma \text{Pb} & \\
\text{Pb @ } 599 \text{ deg} = \sigma \text{Pb} & \\
(1\text{g}, \sigma \text{Pb @ } 600 \text{ deg} + 5.86 \text{ cal}, \Delta) = (1\text{g}, \lambda \text{Pb @ } 600 \text{ deg}) & \\
\text{and so on} & (4.3)
\end{array}
$$

The elements Ag and Pb are known. Our first few lines simply add the "adjective" σ to these symbols and we state that Ag is σ Ag at various temperatures. However, when a certain temperature is reached the addition of more Δ to Ag results in a change from σ to λ Ag with no increase in temperature. It seems that anyone who has tentatively identified Δ with heat and Temp with temperature would conclude that σ Ag and λ Ag are two states of Ag. The melting behaviour of solids is so characteristic that 4.3 should be unmistakable. If so, then since the melting points of Ag and Pb should be independently known by the aliens, and since they know our gram they can use 4.3. to find our calorie and degree without using the information given earlier.

We used metals in 4.3 because their melting points are not greatly affected by pressure. We turn now to a discussion of pressure.

$$
\begin{array}{ll}
Q, \text{Cmat},, \text{Press} \in \text{Fun}(\text{Cmat}, Q) & \\
(L \in \text{Cmat}) \wedge (\text{Press}(L) = 1 \text{ atm}) \leftrightarrow (L \text{ @ } 1 \text{ atm}) \text{ and so on.} & (4.4)
\end{array}
$$

This line introduces the pressure function and its associated terminology. The notion of critical pressure can now be used to make the meaning of 4.4 clear and to provide another means of finding our degree. This redundancy is intentional.

$$
\begin{array}{ll}
(t > 405) \rightarrow (\forall p)((\text{NH}_3 \text{ @ } t \text{ deg @ } p \text{ atm}) = \gamma \text{NH}_3) & \\
(t \leq 405) \rightarrow (\exists p)((\text{NH}_3 \text{ @ } t \text{ deg @ } p \text{ atm}) = \lambda \text{NH}_3) & \\
(t > 5) \rightarrow (\forall p)((\text{He @ } t \text{ deg @ } p \text{ atm}) = \gamma \text{He}) & \\
(t \leq 5) \rightarrow (\exists p)((\text{He @ } t \text{ deg @ } p \text{ atm}) = \lambda \text{He}) & (4.5)
\end{array}
$$

The fact that we know the critical temperature of He says something about our technological competence. We now use the concept of critical pressure to fix (i.e. communicate) the value of 1 atm.

$$
\begin{array}{ll}
(p > 112) \rightarrow ((\text{NH}_3 \text{ @ } 405 \text{ deg @ } p \text{ atm}) = \lambda \text{NH}_3) & \\
(p < 112) \rightarrow ((\text{NH @ } 405 \text{ deg @ } p \text{ atm}) = \gamma \text{NH}_3) & \\
(\text{NH}_3 \text{ @ } 405 \text{ deg @ } 112 \text{ atm}) = (\gamma \text{NH}_3) \cup (\lambda \text{NH}_3) & (4.6)
\end{array}
$$

We can repeat 4.6 for He which has a critical pressure of 2.26 atm. From this information our unit of pressure can be found. We now give some useful, and probably known, facts about water.

$$
\begin{array}{ll}
(1\text{g}, \sigma\text{H}_2\text{O @ } 273 \text{ deg @ } 1 \text{ atm} + 79.7 \text{ cal}. \Delta) \rightarrow & \\
(1\text{g}, \lambda\text{H}_2\text{O @ } 273 \text{ deg @ } 1 \text{ atm}) & \\
(1\text{g}, \lambda\text{H}_2\text{O @ } 373 \text{ deg @ } 1 \text{ atm} + 539 \text{ cal}, \Delta) \rightarrow & \\
(1\text{g}, \gamma\text{H}_2\text{O @ } 373 \text{ deg @ } 1 \text{ atm}). & (4.7)
\end{array}
$$

We have discussed two of the three variables necessary to describe the gaseous state. The third one is volume. It is not easy to give a general definition of volume. However, so many common phenomena, besides the study of gases (e.g. the expansion of liquids when heated, density, etc.), involve this concept that a society with a technology will have come to terms with this notion. So we shall take the approach of assuming that the aliens understand volume and that the task is to make them realise that this, familiar, concept is what we are trying to communicate. We start with the properties of liquid water.

$$
\begin{array}{ll}
L = 1\text{g}, \lambda\text{H}_2\text{O},, \text{vol}(L \text{ @ } 277 \text{ deg}) = 1 \text{ cm}^3 & \\
\text{vol}(L \text{ @ } 277 \text{ deg}) = 1 \text{ cm}^3,, \text{vol}(L \text{ @ } 273 \text{ deg}) = 1.0013 \text{ cm}^3 & \\
\text{vol}(L \text{ @ } 277 \text{ deg}) = 1 \text{ cm}^3,, \text{vol}(L \text{ @ } 275 \text{ deg}) = 1.0003 \text{ cm}^3 & \\
\text{vol}(L \text{ @ } 277 \text{ deg}) = 1 \text{ cm}^3,, \text{vol}(L \text{ @ } 279 \text{ deg}) = 1.0003 \text{ cm}^3 & \\
\text{vol}(L \text{ @ } 277 \text{ deg}) = 1 \text{ cm}^3,, \text{vol}(L \text{ @ } 283 \text{ deg}) = 1.0027 \text{ cm}^3 & \\
1000 \text{ cm}^3 = 1 \text{ liter} & (4.8)
\end{array}
$$

We keep stressing the definition of 1 cm^3 by repeating it in 4.8. The behaviour of water near its freezing point and the importance of this behaviour aquatic life is well known to us and, perhaps, to other life forms as well.

Now we state, very briefly, the gas laws.

vol (2g, H_2 @ 273 deg @ 1 atm) = 22.4 liter
vol (16g, CH_4 @ 273 deg @ 1 atm) = 22.4 liter
vol (17g, NH_3 @ 273 deg @ 1 atm) = 22.4 liter
vol (4g, He @ 273 deg @ 1 atm) = 22.4 liter
vol (4g, He @ 273 deg @ 2 atm) = 11.2 liter
vol (4g, He @ 546 deg @ 1 atm) = 44.8 liter and so on. (4.9)

Finally, we shall give the ideal gas equation for the case of He and the value of the proportionality constant. If our units of pressure and temperature have been understood the aliens can use the value of this constant to compute our unit for volume independently of our gram. Once that is done the gram can be recalculated from our definition of 1 cm^3. This gives another way of finding our gram.

L = 4g, He,,press (L) x vol (L) = R x Temp (L)
R = 0.8027 and so on (4.10)

7. STAGE 5 - MORE MATHEMATICS

It has often been suggested that any scientific society would know about the number π and its relationship to the geometric properties of the circle and the sphere. A circle of radius r has a circumference of length $2\pi r$ and an area of πr^2, and a sphere of radius r has a surface area of $4\pi r^2$, and a volume of $\frac{4}{3}\pi r^3$. It is conceivable that these formulas could be arrived at empirically and that some approximation to π, like $\frac{22}{7}$, could be found the same way. However, it seems reasonable that these formulas themselves would lead a scientifically-oriented society to a deeper study of the ideas underlying them; i.e., to an investigation of the mathematical model behind the physical reality. It is illuminating to recall how we humans deal with these matters, the point of the discussion is that even a "simple" thing like volume leads to the integral calculus.

For the sake of definiteness let us discuss how one defines the volume of a sphere. We begin by assigning to a cube with sides of length s a volume of s^3 cubic units. This establishes our unit of volume. Next we fill our sphere with cubes each having the same size, add the volumes of these cubes and the number we obtain is an approximation to the "volume" of the sphere. It is only an approximation because the "flat" cubes cannot fit into every portion of the "round" sphere. In order to improve this approximation we fill the sphere with smaller cubes, all of the same size. These smaller cubes fit better into the sphere and so the sum of their volumes is a better approximation to the "volume" of the sphere. We continue in this way. At each step we take smaller cubes thereby getting a better "fit" and hence a better approximation to the "volume" of the sphere. Finally, we define the volume of the sphere to be the limit of the numbers obtained in this way. This limit can be computed using the methods of the integral calculus and, for a sphere of radius r, is $\frac{4}{3}\pi r^3$. The important point here is that when one tries to define the volume of a sphere, the area of a circle, or even the length of the circumference of a circle, one is led to consider limiting processes.

Many physical concepts, like instantaneous velocity or acceleration, involve limits. These are fairly sophisticated concepts but, as we indicated above, many elementary problems also lead to limits and hence the ideas of the calculus. There are several things to notice here. First, many people find limits intuitively clear, they think in "continuous" terms and find the notion of better and better approximations leading to the "correct" value for a volume or a velocity obvious. This may tell us something about reality on the "human level," because quantum mechanics tells us that reality, at a certain level, is not "continuous". The second point is that the set Q of rational numbers is inadequate for the purposes of calculus. This is because the limit of a set of rational numbers need not be a rational number. This is why the system of real numbers had to be introduced into mathematics. Roughly speaking it is the smallest number system which contains the rational numbers and has the properety that the limit of any set of real numbers is a real number. So, for example, this is where the number π "lives". It is the real, non-rational, number which is the limit of the set of rational numbers $\frac{314}{100}, \frac{31415}{10{,}000}, \frac{3141592}{10^6}, \frac{314159265}{10^8}, \ldots$. Now let us again stress that many people who know and understand calculus get by very well with an intuitive understanding of the reals. We think of a horizontal line, an arbitrary point which we call 0, an arbitrary unit which we call 1, and we assign the coordinate +1 to the point one unit to the right of 0. We then assign rational coordinates to points in the natural way and those points left without coordinates are assigned irrational numbers. All this is clear enough, but should we assume that any intelligent beings would find the reals so "obvious"? Would they invent/discover the calculus? There are philosophers who believe that mathematical objects, like real numbers, are not just human mental constructs but have an independent existence; perhaps in some meta world. If this is true, then we discovered our mathematics and it is not too hard to believe that others might have made the same discovery [2]. There are, however, also those who believe that our mathematics is a special case. See, for example [7].

The final stage below assumes that the aliens have developed an understanding of real numbers and an approach to the volumes and surface areas of three-dimensional solids equivalent to our own. Constructions of the reals from the rationals were not given until the last century, roughly in the period from 1860-1880, by three Germans, K. Weierstrass, G. Cantor and R. Dedekind. it was also found that there are about half a dozen ways of precisely describing the reals, without constructing them. This is what we shall do [9].

π,,π = 3.14 etc,, π = 3.1415 etc,, π = 3,141592 etc,,π = 3.14159265 etc,,etc
N,,Z,,Q,,$\mathfrak{R}$,,Q $\subseteq \mathfrak{R}$,,$\pi \in \mathfrak{R}$, ~($\pi \in$ Q)) (5.1)

$\mathfrak{R}$,,($\forall$ a $\in \mathfrak{R}$) ($\forall$ b $\in \mathfrak{R}$) (((a + b) $\in \mathfrak{R}$) $\wedge$ ((a x b) $\in \mathfrak{R}$))
($\forall$ a $\in \mathfrak{R}$) ((a + 0 = 0 + a = a) $\wedge$ (a x 1 = 1 x a = a))
($\forall$ a $\in \mathfrak{R}$)($\exists$ (-a) $\in \mathfrak{R}$) (a + (-a) = (-a) + a = 0)
($\forall$a $\in \mathfrak{R}$) (a $\neq$ 0) ($\exists$ $a^{-1} \in \mathfrak{R}$) (a^{-1} x a = a x a^{-1} = 1) (5.2)

$\wp$,, $\wp \subseteq \mathfrak{R}$,, ~ ($\wp = \varnothing$),, ~ ($\wp = \mathfrak{R}$),, $\wp$= { r $\in \mathfrak{R}$ | 0 < r }
($\forall$p$\in$ $\wp$)($\forall$ q $\in \wp$) (((p + Q $\in \wp$) $\wedge$ ((p x q) $\in \wp$)) ,, $-\wp$ =
($\forall$ r$\in \mathfrak{R}$) ((r $\in \wp$) $\vee$ (r $\in -\wp$) $\vee$ (r = 0)) (5.3)

Lines 5.1, 5.2, 5.3 say that $\mathfrak{R}$ is an ordered field containing Q. The fundamental "completeness" property of $\mathfrak{R}$, not shared by Q, is described in 5.4 below using Dedekind cuts [9].

$(\Re - \mathrm{cut}(S,T)) \leftrightarrow (\Re = S \cup T) \wedge (S \neq \varnothing \neq T) \wedge (\forall\, s \in S) \wedge (\forall\, t \in T)(s < t)$
$((\forall\, S,T))\ (\Re - \mathrm{cut}\ (S,T))\ (\exists\, c \in \Re)\ ((\ (r < c) \rightarrow r \in S) \wedge\ (c < r \rightarrow r \in T))$

(5.4)

In a lengthier version we might give an example of a cut of Q for which no cut number c exists.

$(\forall\, r \in R)\ ((0 < r) \rightarrow\ (\exists\, u \in R)\ ((0 < u) \wedge (u^2 = r))$
$(\forall\, r\ \in R)\ (0 < r)\ ((\mathrm{sqrt}(r) = u) \leftrightarrow ((0 < u) \wedge (u^2 = r)))$
$\mathrm{sqrt}(0) = 0$ (5.5)

Line 5.5 gives us the useful square root function. Let us define three-dimensional Euclidean space.

$\Re^3\ =\ \Re \times \Re \times \Re,, (\vec{x} \in \Re^3) \leftrightarrow$
$(\vec{x} = (x_1,x_2,x_3)) \wedge (x_1\ \in \Re) \wedge (x_2 \in \Re) \wedge (x_3 \in \Re)$
$(\forall\ \ \vec{x} \in \Re^3)\ (\ \forall\ \vec{y} \in \Re^3)\ \mathrm{dist}((\vec{x},\vec{y}))$
$= \mathrm{sqrt}((x_1 - y_1)^2\ +\ (x_2 - y_2)^2\ +\ (x_3 - y_3)^2)$ (5.6)

$(\forall \vec{a} \in \Re^3)\ (\forall\, r \in \Re)\ (0 \leq r) \left(B(r,\vec{a}) =\ (\vec{x} \in\ \Re^3 |\ \mathrm{dist}(\vec{x},\vec{a}) \leq r) \right)$
$\vec{a} = \mathrm{cent}(B(r,\vec{a})),, r = \mathrm{rad}(B(r,\vec{a}))$,
$(\forall\ \vec{a} \in \Re^3)\ (\ \forall\ \ r\ \in \Re)\ (0 \leq r) \left(S\ (r,\vec{a}) = (\vec{x} \in\ \Re^3 |\ \mathrm{dist}(\vec{x},\vec{a}) =\ r \right)$
$\vec{a} = \mathrm{cent}\ (S(r,\vec{a})),, r = \mathrm{rad}\ (S(r,\vec{a})),, S(r, \vec{a}) = \mathrm{surf}\ B(r,\vec{a})$.

(5.7)

Line 5.7 contains the definitions of a ball and a sphere of radius r, centre $\vec{a}$. We also note that the surface of a ball is a sphere.

$1\ \mathrm{cu} = \{\vec{x} \in \Re^3 | (\vec{x} = (x_1,x_2,x_3))$
$\wedge\ (0 \leq\ x_1 \leq 1) \wedge (0 \leq\ x_2 \leq 1) \wedge (0 \leq\ x_3 \leq\ 1)\}$
$\mathrm{vol}\ (1\ \mathrm{cu}) = 1, (\forall \vec{a} \in \Re^3)\ (\forall r \in \Re)\ (0 \leq r)\ (\mathrm{vol}\ B(r,\vec{a}) = \frac{4}{3}\ x\ \pi\ x\ r^3)$
$1\mathrm{sq} = \{\vec{x} \in \Re^3 |\ (\vec{x} = (x_1,x_2,x_3)) \wedge (0 \leq\ x_1 \leq\ 1)\ \ (0 \leq\ x_2 \leq\ 1) \wedge (x_3 = 0)\}$

$\mathrm{area}\ (1\ \mathrm{sq}) = 1,$

$(\forall \vec{a} \in \Re^3)\ \ (\ \forall r \in \Re)\ (0 \leq\ r)\ (\mathrm{area}\ S(r,\vec{a}) =\ 4\ x\ \pi\ x\ r^2)$

(5.8)

Finally, we make the connection between our mathematical model just given and the "real-world" volume done in Stage 4.

$(B(r,\vec{a}) \in \mathrm{Cmat}) \wedge (\mathrm{rad}\ B(r,\vec{a}) = r\ \mathrm{cm}) \rightarrow (\mathrm{vol}\ B(r,\vec{a}) = \frac{4}{3}\ x\ \pi\ x\ r^3,\ \mathrm{cm}^3)$
$1\mathrm{cm}^3 = 1\ \mathrm{cm}\ x\ 1\ \mathrm{cm}\ x\ 1\ \mathrm{cm}$

$(B(r\vec{a}) \in \mathrm{Cmat}) \wedge (\mathrm{rad}\, B(r,\vec{a}) = r\ \mathrm{cm}) \rightarrow \mathrm{area}\ (\mathrm{surf}\, B(r,\vec{a})) = 4\ x\ \pi\ x\ r^2,\ \mathrm{cm}^2)$

$1\ \mathrm{cm}^2 = 1\ \mathrm{cm}\ x\ 1\ \mathrm{cm}.$ (5.9)

8. CONCLUSION

There are now many things to converse about, such as the Earth and its atmosphere and the Sun and solar system. Stage 5 included enough mathematics to begin discussing analytic geometry and so if the contact was with a race that possesses the sense of sight, we can easily develop the capability to send them detailed pictures. Since we have done both pressure (4.4) and area (5.8) we can actually define force and hence being discussing mechanics. Celestial mechanics, in particular, would probably be "universally" known, but here again we must either develop the calculus, or assume it to be known. It is interesting to note that the digital computer has no understanding of limits or continuity or any of the central notions of analysis. To some this is the most compelling "proof" that such machines are unintelligent. However that may be, a great deal of analysis can be aproximated by discrete models and these, in turn, can be translated into the language of the computer in such a way that the machine can provide workable answers to real-world problems. This is what much of numerical analysis is about. It may be that some intelligence somewhere might develop a purely discrete view of reality and never even conceive of continuous mathematics. We are thinking here of a truly intelligent computer-like being. Of course, our statement that a purely discrete view of reality may be possible immediately suggests that there might also be an intelligence with a purely continuous view of nature. This cannot be ruled out either.

The question arises, now that we have some idea of what we could say, as to what we should say. This is a matter that deserves serious attention and should be widely discussed for many reasons, one of them being that a consensus on this question would help guide future work on languages like ours.

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